

Comparative Analysis of CNN and Transformer Architectures for Multi-Class Vertebral Segmentation in CT Imaging

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Abstract—Accurate vertebral segmentation from computed tomography (CT) scans is a prerequisite for computer-aided diagnosis, surgical planning, and biomechanical analysis of the spine. This work presents a systematic comparison of three state-of-the-art deep learning architectures—nnUNet (convolutional), MedNeXt (ConvNeXt-inspired convolutional), and SwinUNETR (Transformer-based)—on the VerSe 2019+2020 benchmark, a large-scale multi-class dataset encompassing 26 vertebral labels across 374 CT volumes. All models were trained and evaluated under an identical 3D low-resolution experimental protocol using the nnU-Net self-configuring framework to ensure fair comparison. MedNeXt achieved the best overall performance with a mean Dice Similarity Coefficient (DSC) of 0.854 and an identification rate of 91.5%, outperforming nnUNet (DSC = 0.833, ID = 88.4%) and SwinUNETR (DSC = 0.758, ID = 79.3%). Geometric surface analysis confirms that CNN-based models produce more geometrically consistent reconstructions, with MedNeXt achieving the lowest Hausdorff distance (25.03 mm) and volume error (3.66%). These results indicate that models combining CNN inductive biases with modern architectural design choices—such as the ConvNeXt macro-design adopted by MedNeXt—outperform pure Transformer architectures in data-limited medical imaging scenarios, where self-attention encoders lack sufficient training signal to compensate for their reduced inductive bias.

Index Terms—vertebral segmentation, CT imaging, nnUNet, MedNeXt, SwinUNETR, deep learning, medical image segmentation, VerSe

I. INTRODUCTION

Spinal pathologies—including degenerative disc disease, scoliosis, fractures, and metastatic lesions—affect hundreds of millions of people worldwide and represent a leading cause of disability [1]. Computed tomography (CT) remains the gold standard for imaging the osseous structures of the spine due to its high bone contrast and clinical availability. Accurate, automated segmentation of individual vertebrae is a fundamental building block for downstream clinical tasks such as radiation therapy planning, pedicle screw placement guidance, vertebral body morphometry, and population-scale epidemiological studies.

Manual vertebral delineation is, however, time-consuming and subject to inter-rater variability. A single CT volume

may contain up to 28 vertebral bodies spanning cervical (C1–C7), thoracic (T1–T12), lumbar (L1–L5), and sacral (S1) levels, each of which must be individually labeled in three dimensions. Automating this process at scale requires models capable of handling the full anatomical variability of the spine—including degenerative changes, post-surgical implants, and truncated fields of view—while maintaining fine-grained per-vertebra identification.

The deep learning revolution has produced a rich ecosystem of segmentation architectures. Convolutional neural networks (CNNs), in particular the U-Net family [2], have dominated medical image segmentation for nearly a decade owing to their strong inductive biases for local feature extraction and translation equivariance. The nnUNet framework [4] unified this family by introducing self-configuring preprocessing and architecture selection, becoming a competitive baseline on virtually every medical segmentation benchmark.

In parallel, Vision Transformers (ViTs) [8] have demonstrated that attention-based global context modeling can capture long-range dependencies that convolutions struggle with. SwinUNETR [6] adapts the hierarchical Swin Transformer [7] as an encoder within a U-Net decoder to combine global context with multi-scale feature fusion. More recently, MedNeXt [5] revisited the convolutional paradigm by adopting the ConvNeXt [9] macro-design in 3D—depthwise separable convolutions, inverted bottleneck blocks, layer normalization, and GELU activations—improving representation quality without the quadratic complexity of self-attention.

Despite the abundance of proposed architectures, rigorous, controlled comparisons on vertebral segmentation datasets remain limited. Most published comparisons differ in preprocessing, data splits, or evaluation protocols, making it difficult to draw actionable conclusions. In this work, we address this gap by evaluating nnUNet, MedNeXt, and SwinUNETR under an identical experimental protocol on the VerSe 2019+2020 dataset [11], the largest publicly available benchmark for vertebral CT segmentation.

Our main contributions are:

- A controlled, fair comparison of three representative architectures spanning the CNN–Transformer spectrum.
- An extended evaluation including both voxel-level metrics (DSC, RMSD, identification rate) and surface-level geometric metrics (ASSD, Chamfer, HD95, Hausdorff, volumetric error).
- Analysis of failure modes with qualitative visualizations and discussion of architectural implications for spine imaging.

II. RELATED WORK

A. Convolutional Architectures for Vertebral Segmentation

The U-Net [2], introduced by Ronneberger et al. for 2D biomedical image segmentation, established the encoder-decoder paradigm with skip connections that has since been extended to 3D volumetric data [3]. For spinal CT, CNN-based methods have demonstrated strong performance: Payer et al. [12] combined a detection network with a local segmentation network to handle the multi-label nature of the spine, while Löffler et al. [13] leveraged a graph-based instance detection followed by U-Net segmentation. The nnUNet framework [4] by Isensee et al. generalizes these insights into a self-configuring pipeline that automatically determines architecture depth, patch size, and normalization based on dataset fingerprints, consistently ranking among the top methods on medical segmentation benchmarks.

B. Transformer-Based Medical Image Segmentation

Following the success of ViT [8] in natural image classification, several works adapted self-attention mechanisms for medical image segmentation. TransUNet [10] serialized CNN feature maps and processed them with Transformer encoders before decoding with U-Net skip connections. SwinUNETR [6] replaced the encoder entirely with a Swin Transformer [7], whose shifted window attention enables hierarchical, computationally efficient feature extraction from 3D volumes. The model achieved state-of-the-art results on the Medical Segmentation Decathlon [15] and has been widely adopted in subsequent works.

C. Large-Kernel Convolutional Approaches

ConvNeXt [9] demonstrated that modernized CNNs with large depthwise convolutions can match or exceed Transformer performance on image classification. MedNeXt [5] by Roy et al. extended this concept to 3D medical image segmentation, replacing standard $3\times 3\times 3$ kernels with larger $5\times 5\times 5$ to $7\times 7\times 7$ kernels and adopting residual ConvNeXt blocks. By integrating within the nnU-Net framework, MedNeXt maintains automated configuration while enlarging the effective receptive field, yielding gains especially in tasks requiring context over larger regions such as multi-vertebra delineation.

D. VerSe Challenge

The Vertebra Segmentation (VerSe) challenge [11] by Sekuboyina et al. consolidated the VerSe 2019 and VerSe 2020 editions, providing 374 CT scans with per-vertebra annotations

across 28 classes from multiple clinical sites. The challenge introduced standardized evaluation metrics including DSC, identification rate, and root-mean-square distance (RMSD), enabling reproducible comparisons. Published top performers on VerSe include iterative refinement pipelines and cascade architectures, but few papers directly compare CNN and Transformer models within the nnUNet framework.

III. METHODOLOGY

A. Dataset

We use the VerSe 2019+2020 combined dataset (Dataset507_VerSe2020) [11], comprising 374 CT volumes from 355 unique patients. Vertebrae are annotated with 26 instance labels observed across the dataset: C1–C7 (7 cervical), T1–T12 (12 thoracic), L1–L5 (5 lumbar), and S1 (1 sacral), with T13 as an additional transitional anatomical variant (label index 27) present in a subset of cases. The VerSe label convention reserves up to 28 class indices; label indices 26 and 28 are absent from this dataset split. The dataset is split into 261 training cases and 113 test cases following the official VerSe partition. Images are stored in NIfTI format (.nii.gz) with varying fields of view, resolutions, and patient orientations, reflecting the distribution heterogeneity typical of multi-center clinical collections. The reference paper by Sekuboyina et al. [11] provides a detailed description of the annotation protocol and scanner diversity.

B. Preprocessing

All three models are trained and evaluated using the nnUNet preprocessing pipeline [4] configured for the `3d_lowres` plan. The pipeline automatically resamples volumes to a common isotropic spacing of approximately 1.8 mm (median: $1.81\times 1.76\times 1.76$ mm), normalizes CT intensities using global foreground statistics (CT normalization), and crops to non-zero bounding boxes. Training patches of size $128\times 128\times 128$ voxels are extracted with a batch size of 2. Data augmentation includes random rotation, scaling, mirroring, elastic deformations, Gaussian noise, and brightness jitter—all applied on-the-fly during training.

Class imbalance. The VerSe 2019+2020 dataset exhibits inherent class imbalance: thoracic vertebrae (T1–T12) are present in virtually all volumes, whereas cervical (C1–C7) and upper sacral (S1) labels appear in fewer scans and occupy smaller volumes. To address this, nnUNet’s default sampler enforces that at least 33% of patches in each training batch contain foreground voxels, drawn with uniform probability across all annotated classes. This foreground oversampling prevents the loss from being dominated by the majority thoracic classes and ensures adequate gradient signal for rarer cervical and sacral labels. All three models benefit from this strategy, as they share the same nnUNet data loader.

C. Model Architectures

1) *nnUNet (PlainConvUNet)*: The baseline model is nnUNet’s automatically configured 3D PlainConvUNet, consisting of 6 encoder stages with feature channels

[32, 64, 128, 256, 320, 320], 3D $3 \times 3 \times 3$ convolutional blocks, instance normalization [14], and LeakyReLU activations. Downsampling is performed via strided convolution ($2 \times 2 \times 2$) at each stage except the first. The symmetric decoder uses bilinear upsampling followed by convolutions, with skip connections from the encoder. The final layer projects to 28 class logits.

2) *MedNeXt*: MedNeXt [5] replaces the standard convolutional blocks with residual ConvNeXt-inspired blocks. We use the MedNeXt-B (base) variant with `kernel_size=3`, retaining $3 \times 3 \times 3$ spatial convolutions but adopting the ConvNeXt macro-design: depthwise separable convolutions, an inverted bottleneck structure, layer normalization, and GELU activations. The architecture retains the same U-Net topology and nnUNet preprocessing as the baseline.

3) *SwinUNETR*: SwinUNETR [6] replaces the convolutional encoder entirely with a Swin Transformer, processing the input volume as a sequence of non-overlapping 3D tokens. Shifted window self-attention [7] enables efficient local-global feature modeling. The hierarchical encoder outputs are fed into a 3D convolutional decoder via skip connections, following the standard UNet design. The same nnUNet preprocessing and data splits are applied, with the patch size and spacing identical to the other two models.

D. Training Protocol

All models are trained using 5-fold cross-validation on the 261 training cases, with the final model obtained by ensemble averaging of the 5 folds. nnUNet uses its default SGD optimizer (Nesterov momentum 0.99, initial learning rate 0.01, polynomial decay). MedNeXt and SwinUNETR use AdamW with initial learning rates of 10^{-3} and 10^{-4} respectively, both with CosineAnnealingLR decay to 10^{-6} over 250 epochs. All models are trained with a combined Dice + Cross-Entropy loss for a maximum of 250 epochs per fold. All experiments were conducted on a single NVIDIA RTX 3090 24 GB GPU provisioned via the RunPod.io cloud platform. Per-fold GPU times were 3.57 h (nnUNet), 7.59 h (MedNeXt), and 16.19 h (SwinUNETR); total 5-fold training required 17.85 h, 37.95 h, and 80.95 h respectively. The best validation Dice recorded during training was 0.851 for nnUNet. Per-epoch pseudo-Dice values logged during training are not directly comparable across models, as each trainer’s validation loop uses a different approximation; final performance is evaluated on the held-out test set and reported in Table I.

E. Evaluation Metrics

We report the following metrics on the 113 official VerSe test cases:

Voxel-level metrics:

- **DSC / F1-score** (Dice Similarity Coefficient): measures volumetric overlap between predicted and ground-truth segmentations, computed per vertebra and averaged across all 26 observed vertebral labels. DSC is mathematically equivalent to the F1-score in binary segmentation

($\text{DSC} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$); multi-class F1 is therefore reported as the mean DSC across all 26 observed vertebral labels.

- **ID Rate** (Identification Rate): fraction of vertebrae correctly identified (i.e., predicted instances matched to a ground-truth vertebra with $\text{DSC} > 0$).
- **RMSD** (Root Mean Square Distance): average root mean square distance between centroids of predicted and matched ground-truth vertebrae, reported in mm.

Surface-level geometric metrics (computed on 3D mesh reconstructions):

- **ASSD**: Average Symmetric Surface Distance.
- **Chamfer**: Chamfer distance between surface point clouds.
- **HD95**: 95th-percentile Hausdorff distance.
- **Hausdorff**: Maximum directed Hausdorff distance.
- **Vol. Error (%)**: percentage error in predicted vertebral volume.

IV. EXPERIMENTAL RESULTS

A. Voxel-Level Segmentation Performance

Table I summarizes the mean voxel-level metrics across the 113 test cases. MedNeXt consistently outperforms both alternatives in DSC (+0.021 over nnUNet, +0.096 over SwinUNETR), identification rate (+3.1% over nnUNet, +12.2% over SwinUNETR), and RMSD (2.63 mm vs. 2.96 mm and 4.30 mm, respectively). nnUNet ranks second across all metrics, while SwinUNETR shows notably weaker performance, particularly in identification rate (79.3%) suggesting more frequent complete misses of individual vertebrae.

TABLE I
MEAN VOXEL-LEVEL EVALUATION METRICS ON VERSE TEST SET
(N = 113)

Model	DSC $\uparrow$	ID Rate $\uparrow$	RMSD (mm) $\downarrow$
nnUNet	0.833	0.884	2.96
MedNeXt	0.854	0.915	2.63
SwinUNETR	0.758	0.793	4.30

Per-case analysis reveals that all three models share common hard cases (e.g., cases 0032, 0088, 0099) involving severe deformity or extreme truncation of the field of view, where even nnUNet achieves $\text{DSC} < 0.2$. MedNeXt recovers more gracefully on borderline cases, while SwinUNETR exhibits higher variance with several additional low-performing cases not observed in the CNN-based models.

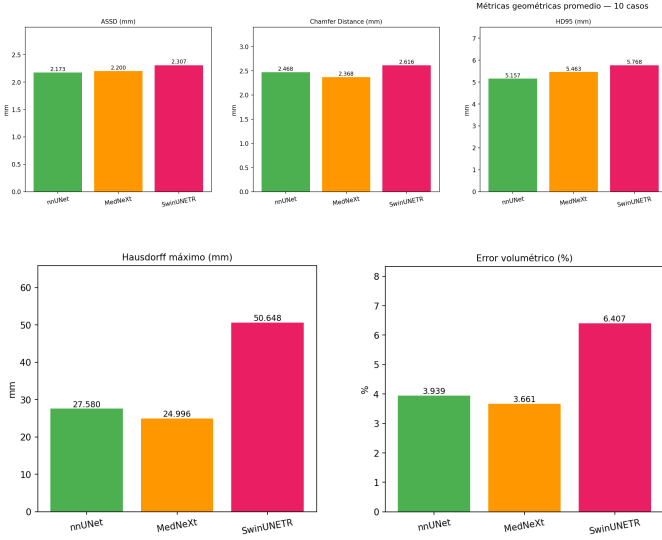


Fig. 1. Geometric surface metrics for the three models averaged over the test set (10 sampled cases). Top row: ASD, Chamfer distance, and HD95. Bottom row: Hausdorff distance and volumetric error (%).

B. Geometric Surface Metrics

Table II reports surface-level geometric metrics computed from 3D mesh reconstructions of the segmentation outputs. Due to the high computational cost of mesh extraction and point-cloud comparison, these metrics were evaluated on a randomly sampled subset of 10 test cases (cases 0004, 0014, 0015, 0018, 0029, 0032, 0036, 0082, 0087, 0095). MedNeXt achieves the best Chamfer distance (2.37 mm), Hausdorff distance (25.03 mm), and volumetric error (3.66%), indicating superior fidelity of the predicted surface geometry. nnUNet obtains the best HD95 (5.22 mm), reflecting fewer moderate-sized boundary errors, while MedNeXt and nnUNet are comparable in ASD (2.20 vs. 2.17 mm). SwinUNETR records substantially higher Hausdorff distance (50.92 mm) and volumetric error (6.41%), confirming the voxel-level trend of more frequent large outlier predictions.

TABLE II
MEAN GEOMETRIC SURFACE METRICS ON VERSE TEST SET

Model	ASDD (mm)↓	Chamfer (mm)↓	HD95 (mm)↓	Hausd. (mm)↓	Vol. Err. (%)↓
nnUNet	2.17	2.48	5.22	27.10	3.94
MedNeXt	2.20	2.37	5.44	25.03	3.66
SwinUNETR	2.30	2.61	5.64	50.92	6.41

C. Qualitative Results

Fig. 2 shows a representative CT slice with multi-class vertebral segmentation produced by the three models. The nnUNet and MedNeXt predictions closely follow the ground-truth annotation, with well-defined boundaries at the endplates and pedicles. The SwinUNETR prediction exhibits fragmented labels in the thoracic region, consistent with the lower identification rate reported quantitatively.

VerSe_test_0004

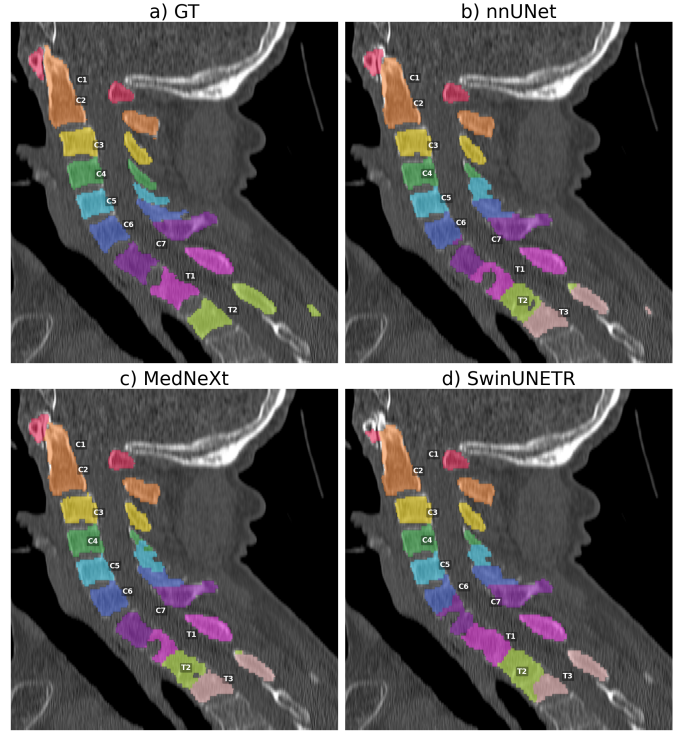


Fig. 2. Representative axial CT slice overlaid with multi-class vertebral segmentation masks. Each color corresponds to a distinct vertebral label (C1–S1).

Fig. 3 presents 3D mesh reconstructions of predicted vertebral columns across selected test cases. MedNeXt and nnUNet produce anatomically plausible meshes with consistent vertebral shapes and inter-vertebral spacing, while SwinUNETR reconstructions show topological inconsistencies in challenging cases.

VerSe_test_0004

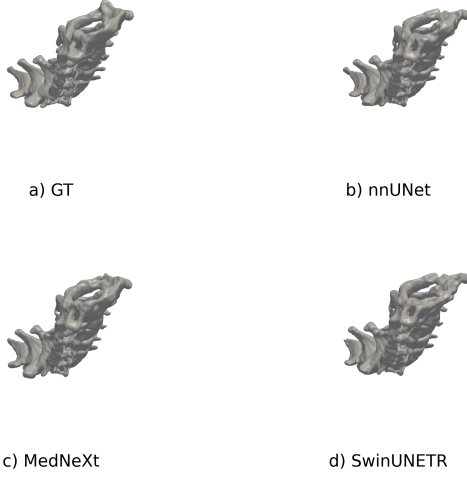


Fig. 3. 3D mesh reconstructions of predicted vertebral columns for representative VerSe test cases. Columns correspond to (from left) nnUNet, MedNeXt, and SwinUNETR predictions.

D. Training Dynamics

The per-fold training curves (learning rate, training loss, validation loss, and EMA pseudo-Dice) over 250 epochs for all three models are shown in Figs. 4–6, logged via Weights & Biases.

nnUNet (Fig. 4) converges slowly but with exceptional stability: all five folds are nearly indistinguishable throughout training, reflecting the robustness of polynomial LR decay and the strong spatial inductive biases of the PlainConvUNet. The EMA pseudo-Dice rises steadily to ≈ 0.80 and remains consistent across folds.

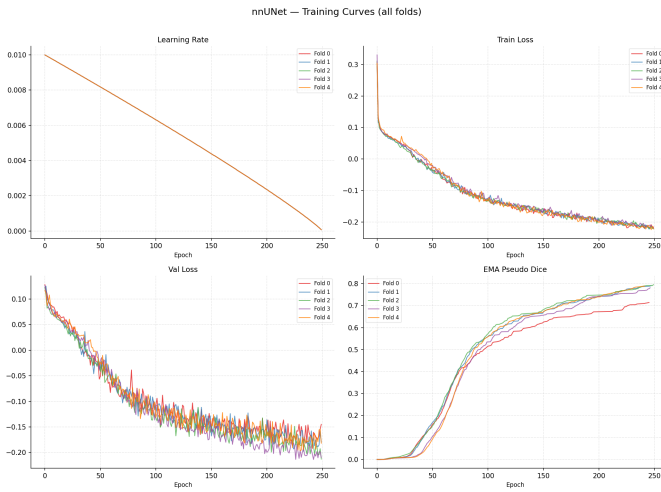


Fig. 4. nnUNet per-fold training curves (5-fold CV). All five folds are nearly indistinguishable, reflecting exceptional stability under polynomial LR decay.

MedNeXt (Fig. 5) reaches its convergence plateau much

faster, stabilizing within ≈ 50 epochs. It achieves the highest final EMA pseudo-Dice (≈ 0.80 – 0.85) among the three models. Fold 0 settles slightly below the remaining folds, consistent with it being the most challenging partition. The AdamW optimizer drives a steeper initial loss descent, while CosineAnnealingLR maintains stable fine-tuning through epoch 250.

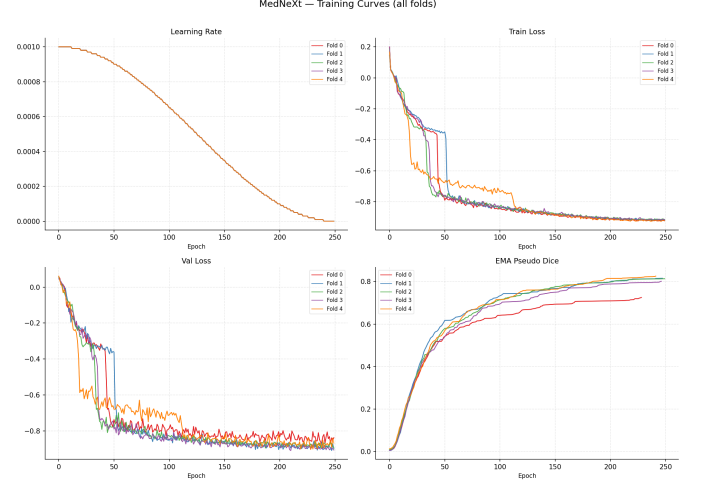


Fig. 5. MedNeXt per-fold training curves (5-fold CV). Convergence plateau reached within ≈ 50 epochs; EMA pseudo-Dice achieves the highest final values among the three models.

SwinUNETR (Fig. 6) never reaches the stable plateau observed in the CNN-based models. The EMA pseudo-Dice remains noisy and oscillating throughout training—an unambiguous visual signature of a data-hungry architecture failing to form stable representations from 261 training volumes. This fold-level instability directly explains the lower and more variable test-set performance in Table I.

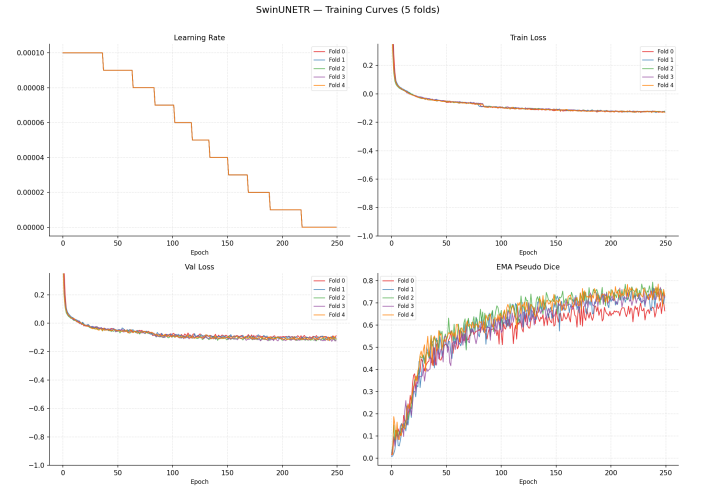


Fig. 6. SwinUNETR per-fold training curves (5-fold CV). EMA pseudo-Dice remains noisy and oscillating throughout all 250 epochs, consistent with insufficient training data for a Transformer encoder.

V. DISCUSSION

A. Why Does MedNeXt Outperform nnUNet?

The margin between MedNeXt and nnUNet is modest (+2.1% DSC) but consistent across all metrics. Both models use $3 \times 3 \times 3$ convolutions (MedNeXt-B with `kernel_size=3`), so the advantage does not come from a larger receptive field. Instead, the gains stem from the ConvNeXt macro-design adopted by MedNeXt: depthwise separable convolutions decouple channel and spatial mixing, the inverted bottleneck structure increases the effective feature dimensionality in the intermediate layers, layer normalization provides more stable gradients than instance normalization, and GELU activations smooth the optimization landscape relative to LeakyReLU. Together, these design choices improve gradient flow and feature representation quality, yielding better generalization from the same 261 training cases without any increase in spatial kernel size.

B. Why Does SwinUNETR Underperform?

The Swin Transformer encoder achieves excellent results when trained on large datasets but relies on large amounts of data to learn competitive representations due to its reduced inductive bias relative to CNNs [8]. The VerSe training set of 261 cases is considered small-to-medium in the context of modern deep learning. Additionally, the *3d_lowres* configuration used here provides patch inputs of 128^3 voxels at ~ 1.8 mm spacing, which corresponds to a relatively coarse spatial resolution—arguably reducing the advantage of fine-grained attention maps. Experiments with higher-resolution inputs or pre-training on larger datasets (e.g., TotalSegmentator [16]) may close this gap.

The low identification rate (79.3%) of SwinUNETR further suggests that the model occasionally fails to detect vertebrae entirely, rather than producing poorly overlapping but topologically correct predictions. This failure mode is more costly in clinical applications where a complete census of the vertebral column is required.

C. Computational Cost

Table III reports the measured computational footprint of each model under the 3D low-resolution configuration on the RTX 3090 24 GB (batch size 2, patch 128^3 , 5-fold training).

TABLE III
MEASURED COMPUTATIONAL COST PER MODEL (RTX 3090 24 GB, RUNPOD.IO, 3D LOW-RES)

Model	Params (M)	Weights (MB)	VRAM (%)	Epoch (s/fold h)
nnUNet	60.59	354.6	38.46	51.5 / 3.57
MedNeXt	31.66	126.6	99.34	109.3 / 7.59
SwinUNETR	63.13	252.5	79.74	216.7 / 16.19

Several counter-intuitive observations emerge. First, MedNeXt has the *fewest* parameters (31.66 M) yet consumes

the most VRAM (99.34% of the 24 GB device). The ConvNeXt macro-design—depthwise separable convolutions, inverted bottlenecks, and layer normalization—does not add many learnable weights but produces larger intermediate activation tensors during the forward and backward passes, driving up peak memory. Second, SwinUNETR’s self-attention mechanism is by far the slowest operator: at 216.7 s per epoch it trains $4.2\times$ slower than nnUNet, accumulating over 80 hours of total GPU time across 5 folds. Third, nnUNet’s checkpoint (354.6 MB) is the largest of the three despite a parameter count similar to SwinUNETR (252.5 MB), because its dense $3 \times 3 \times 3$ convolutions in the deepest stages store full-rank weight tensors, whereas MedNeXt’s depthwise separable structure yields the most compact weights (126.6 MB) with fewer parameters overall.

Taken together, nnUNet offers the best efficiency–accuracy trade-off for resource-constrained deployments. MedNeXt achieves the highest segmentation quality at the cost of nearly saturating the GPU memory budget. SwinUNETR provides neither a segmentation nor an efficiency advantage at this dataset scale, making it the least practical choice for the VerSe task without access to pre-trained weights or substantially more training data.

D. Failure Mode Analysis

Inspecting per-case results reveals three distinct failure patterns that illuminate the architectural differences among the models.

Type 1 — Catastrophic failure due to severely truncated field of view. Case 0032 (Fig. 7) achieves DSC=0.104, 0.253, and 0.003 for nnUNet, MedNeXt, and SwinUNETR respectively. The CT volume covers only a narrow abdominal window, leaving the thoracic vertebrae almost entirely outside the field of view. All three models fail to identify the correct set of visible vertebrae, with SwinUNETR producing near-zero overlap. This type of failure is largely dataset-agnostic: even a perfect architecture cannot localise structures outside the image boundary. Case 0088 (all models DSC < 0.41) exhibits the same pattern.

Type 2 — SwinUNETR-specific fragmentation. In several cases (e.g., 0036, 0087) where nnUNet and MedNeXt achieve DSC > 0.7, SwinUNETR collapses to DSC < 0.2 (Fig. 8). Visual inspection reveals that SwinUNETR generates spatially fragmented label maps—small, disconnected islands assigned to multiple different vertebra classes within a single anatomical level. This is consistent with a model that lacks strong spatial continuity priors, relying instead on global token interactions that are not yet well-calibrated with only 261 training volumes. Connected-component post-processing could partially mitigate this but does not address the underlying misclassification.

Type 3 — MedNeXt recovery on ambiguous junctions. Cases 0036, 0040, and 0096 demonstrate a pattern where nnUNet fails (DSC=0.27–0.46) while MedNeXt recovers significantly (DSC=0.71–0.89). These cases share unusual anatomical configurations at transition regions (cervicothoracic or lumbosacral junctions). Even though MedNeXt-B uses the

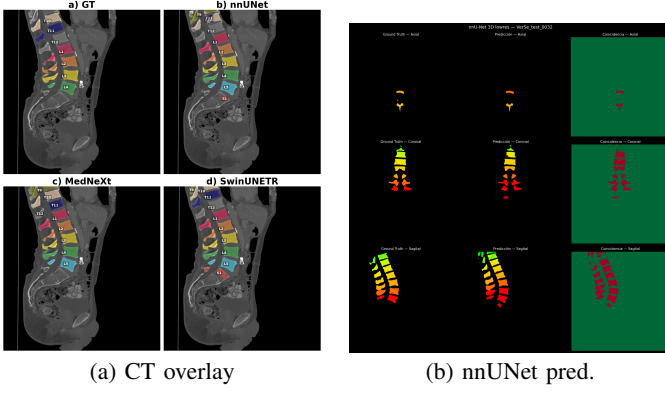


Fig. 7. Case 0032 (Type 1 failure): severely truncated FOV. All models produce near-zero DSC. (a) Ground-truth overlay; (b) nnUNet segmentation showing fragmented, incorrectly identified labels.

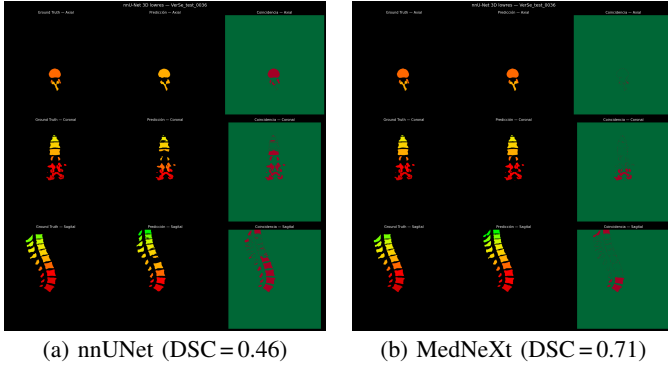


Fig. 8. Case 0036 (Type 2/3 failure): SwinUNETR achieves DSC=0.12 (not shown). (a) nnUNet struggles at the lumbosacral junction; (b) MedNeXt recovers substantially, correctly delineating the L4-S1 region.

same $3 \times 3 \times 3$ kernel size as nnUNet, its ConvNeXt-inspired design—deeper inverted bottlenecks and more expressive feature mixing—improves the model’s ability to distinguish ambiguous vertebral levels from their neighbours, likely through better-calibrated feature representations rather than a wider spatial receptive field.

E. Ensemble Strategies and Hybrid Architectures

An important dimension not explored in this work is the combination of the three models. In theory, an ensemble that averages the softmax outputs of nnUNet, MedNeXt, and SwinUNETR at inference time could yield gains beyond any individual model, since their error distributions differ substantially (Section V-D): SwinUNETR’s unique failures do not systematically overlap with the CNN failures, so a majority-vote ensemble could recover cases that a single model misses. However, the $4.5\times$ training-time overhead of SwinUNETR relative to nnUNet makes this trade-off unattractive at current dataset scales.

It is also worth noting that MedNeXt can itself be interpreted as a *hybrid architecture*: its residual blocks adopt the macro-design principles of Transformers (depthwise convolutions, inverted bottleneck, GELU activations, layer normal-

ization) while retaining the inductive biases of CNNs (local weight sharing, translation equivariance). In this sense, the comparison is not strictly CNN vs. Transformer, but rather a spectrum: nnUNet (pure CNN) $\rightarrow$ MedNeXt (CNN with Transformer-inspired design) $\rightarrow$ SwinUNETR (Transformer encoder). The consistent performance ordering along this spectrum suggests that, at limited training-set sizes, preserving convolutional inductive biases while adopting modern architectural design choices is preferable to full attention-based encoding.

F. Limitations

This study is limited to the 3D low-resolution configuration and does not evaluate cascade architectures, which represent the state-of-the-art for VerSe. Furthermore, no post-processing (e.g., connected-component analysis or anatomical ordering constraints) was applied, which could disproportionately benefit SwinUNETR by correcting isolated fragmented predictions. Future work should also assess model robustness on pathological spines (severe scoliosis, instrumented spines) which are underrepresented in VerSe.

VI. CONCLUSIONS

We presented a controlled comparison of nnUNet, MedNeXt, and SwinUNETR for multi-class vertebral segmentation on the VerSe 2019+2020 dataset under an identical experimental protocol. MedNeXt demonstrated superior overall performance across both voxel-level and geometric surface metrics, attributable to its ConvNeXt-inspired macro-design—depthwise separable convolutions, inverted bottleneck structure, layer normalization, and GELU activations—which improve gradient flow and feature representation without increasing spatial kernel size. nnUNet provides a strong and computationally efficient baseline, while SwinUNETR—despite its theoretical capacity for global context modeling—underperforms in this data-limited setting. Our findings suggest that for small-to-medium medical segmentation datasets, architectures that preserve convolutional inductive biases while adopting modern design principles offer a more favorable trade-off than pure Transformer encoders.

Future directions include: (1) evaluation of cascade architectures at full resolution; (2) self-supervised or semi-supervised pre-training on unlabeled CT datasets to mitigate data scarcity for Transformer models; (3) integration of anatomical priors (vertebra ordering, symmetry) into the loss function; and (4) extension to multimodal data combining CT with MRI for soft-tissue-aware segmentation.

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